

Siva Krishna Pattem

Data Engineer | Databricks | Snowflake

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SUMMARY

Results-driven **Data Engineer** with **6.5+ years** of experience architecting scalable Lakehouse platforms, ELT pipelines, and dimensional models across Databricks, Snowflake, and AWS using **SQL, Python, Pyspark, Airflow and DBT**. Proven record of accomplishment in SAP migrations, automated Medallion frameworks. Certified **Databricks Data Engineer Professional** and **SnowPro Associate**, adept at leveraging AI/LLM tools (**Databricks Genie, OpenAI, Claude and Copilot**) to accelerate metadata workflows and self-service enterprise analytics.

EXPERIENCE

Data Engineer | TCS

Feb 2025 – Present

Client: Global Medical & Healthcare Technology Company

- Architected **conceptual, logical, and physical dimensional models** (Star Schemas) across **10+ subject areas** using **Erwin** to establish standardized enterprise reporting assets.
- Performed deep data discovery and structural analysis across legacy **SAP BW** and **SAP ECC** systems, leveraging **LLM-driven automation** to generate precise **Source-to-Target Mappings (STTM)** and accelerate design workflows.
- Developed and operated production **ELT pipelines** using **SQL, PySpark** and **Apache Airflow** across multi-stage data hubs (**Source --> Curated --> Functional --> Reporting**).
- Engineered an automated, PySpark and Delta Lake **Change Data Feed (CDF)** framework for **Silver** and **Gold Medallion layer** transformations, improving pipeline observability and developer velocity.
- Configured **GitLab CI/CD pipelines** with automated **Unit and SIT testing** across schemas, ensuring seamless data reconciliation and defect-free releases.
- Executed the migration of multi-terabyte datasets from legacy SAP BW to Databricks, enabling the decommissioning of legacy infrastructure while driving 99.9% data availability serving downstream applications.
- Streamlined enterprise data access by configuring **Databricks UniForm Iceberg** properties, establishing **zero-copy integration** to query native Lakehouse storage directly from **Snowflake** for downstream business intelligence.
- Implemented **Databricks Genie** spaces to build interactive dashboards, enable self-service **Natural Language Q&A** over enterprise datasets, and orchestrate agentic actions for business stakeholders.
- Applied advanced performance tuning on complex SQL workloads using **Liquid Clustering, materialized views, and metric views** to significantly reduce query latency and compute consumption.
- Created comprehensive technical data flow diagrams and documentation, while **training and mentoring** new associates on platform best practices.

Analyst – Quality Specialist | Amazon Development Center

May 2022 – May 2024

- Engineered end-to-end serverless ETL pipelines using **AWS S3, Lambda, and Redshift**, automating daily operational workflows and reducing data processing latency by 50%.
- Leveraged advanced SQL window functions (LAG/LEAD), multi-stage CTEs, and funnel analysis on 10M+ clickstream logs to detect critical operational anomalies, safeguarding \$100K+ in daily GMV.
- Architected multi-tier data models in **SQL** and **Python** to track customer and seller lifecycle behaviors, surfacing actionable insights that reduced 90-day churn by 8.5%.
- Built interactive **Tableau** dashboards for WBR leadership reviews and automated audit processes using Excel and Python, saving 15+ manual hours weekly.

Analyst | ICICI Bank

June 2021 – Sept 2021

- Analyzed marketing, credit card, and personal loan datasets to identify key customer insights, directly driving 400+ new account conversions.

Snowflake Developer | Cognizant

June 2018 – April 2021

Client: UK based Telecommunications company

- Engineered scalable **Snowflake** pipelines using **Snowpipe**, **zero-copy cloning**, **time travel**, and custom **UDFs** to streamline real-time ingestion, testing, and system recoverability.
- Built robust **DBT models** and automated orchestrations via **Apache Airflow** to transform raw source data into analytics-ready datasets, standardizing business metrics and eliminating manual pipeline intervention.
- Implemented automated data quality checks within **DBT**, establishing continuous monitoring to catch discrepancies upstream and maintain continuous reliability across business intelligence reporting layers.
- Refactored complex **SQL** queries, procedures, and mapping sessions for large-scale datasets, reducing query execution times, join optimizations, and window functions.
- Developed UNIX shell scripts and automated SQL stored procedures for data reconciliation, saving 10+ operational hours weekly while establishing end-to-end data lineage across source systems.
- Managed production workflows and troubleshooting operations, collaborating directly with clients and business analysts to resolve critical issues and fulfill ad-hoc data requirements.

TECHNICAL SKILLS

AI Tools	Snowflake Cortex, Agent Bricks, OpenAI API, GitHub Copilot, Cursor, Gemini
Databricks	Unity Catalog, Databricks SQL, Lakehouse Architecture, Medallion Architecture, Notebooks, Liquid Clustering, Time Travel, Versioning, Auto Loader, Delta Live Tables (DLT), Lakeflow Jobs, CDC, CDF, AI/BI Genie, DABs, delta sharing.
Snowflake	Snowflake SQL, Architecture, Time Travel, Zero-Copy Cloning, Streams & Tasks, Materialized Views, Secure Data Sharing, Query Profiling, Performance Tuning, Warehouse Sizing, Cost Management, Snowflake Cortex, Snowpark, Real-time & Batch Pipeline Design.
Data Engineering	Apache Airflow (Orchestration), DBT(Data Build Tool), Apache Spark (PySpark / SQL), VS code, draw.io, Apache Iceberg.
Data Modelling	Conceptual Diagrams, Logical/Physical Models, Dimensional Modelling, Star Schema, Snowflake Schema, ERwin, SCD Type 1/2/3, Fact & Dimension Tables, Atlan Data catalog
Programming	SQL, Python, PySpark.
BI & Analytics	Databricks AI/BI Genie, Tableau, Excel, Data Visualization.
DevOps & Tools	Git, GitHub, Gitlab CI/CD Pipelines, Version Control, Agile/Scrum, JIRA

EDUCATION

- Graduation:** B. Tech in Mechanical Engineering (2014-2018) at Bapatla Engineering College, with **7.40/10 CGPA**.
- Intermediate (10+2):** In MPC (2012-2014) at Viveka Junior College, Tenali with **89%**.
- SSC (10th):** (2012) at Z. P. H. School with **9.2/10 GPA**.

CERTIFICATIONS

- Databricks Certified Data Engineer Professional - [link](#)
- Databricks Certified Data Engineer Associate - [link](#)
- SnowPro Associate: Platform Certification - [link](#)
- Google Advanced Data Analytics Certificate - [link](#)